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The Effect of Trunk and Shank Position on the Hip-to-Knee Moment Ratio in a Bilateral Squat

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Abstract

Objectives: The effect of knee position on joint moments during squats has been studied; however, the effect of trunk angle has been less well investigated. This study evaluated the effect of both trunk and knee sagittal plane position on the distribution of moments between the hip and knee extensors during the bilateral squat.

Design: Observational study.

Setting: Biomechanics laboratory.

Participants: One hundred individuals performed bilateral squats.

Main Outcome Measures: Motion and force data were collected using motion capture and force plates. Trunk and shank angles and hip and knee moments were calculated. A linear regression was used to associate the ratio between the hip and knee moments (hip-to-knee moment ratio) with the sagittal plane trunk and shank angles, while accounting for six squat depths (between 60° and 110° of knee flexion) and side.

Results: Trunk angle and shank angle each contributed to the hip-to-knee moment ratio ($P<.001$) with trunk accounting for a higher proportion of variance than the other variables. The hip-to-knee moment ratio increased with increasing trunk angle and with decreasing shank angle.

Conclusions: This large cohort study supports the use of trunk position to instruct squat technique with the goal of modifying hip and knee moments.

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Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Keywords

Biomechanics; trunk angle; shank angle; kinematics; joint moments

1. INTRODUCTION:

The bilateral squat is commonly used in rehabilitation and strength and conditioning settings to improve lower extremity strength and sports performance. The bilateral squat is also used to assess movement patterns, including squat technique, symmetry, lower extremity control, and trunk control (Kritz, Cronin, & Hume, 2009). There has been considerable debate over the “correct” or “best” squat technique. Many individuals advocate for a hip-dominant squat, that better incorporates the hip extensors, or a knee-dominant squat that primarily targets the knee extensors (Figure 1). However, squatting techniques exist on a continuum between hip-dominant and knee-dominant and no one squat type is universally superior to another. Exercise selection should largely rely on the goals of the individual as each squat type has a specific influence on joint kinematics, kinetics, and loading (Chiu, vonGaza, & Jean, 2017, Fry, Smith, & Schilling, 2003, Kernozek et al., 2018, List et al., 2013, Lorenzetti et al., 2012).

Clinicians and coaches often use a verbal or tactile cue directed at knee position to bias an individual’s squat performance to a more hip-dominant (knees behind toes) or knee-dominant (knees in front of toes) movement pattern (Chiu, vonGaza, & Jean, 2017, Fry, Smith, & Schilling, 2003, Kernozek et al., 2018, List et al., 2013, Lorenzetti et al., 2012). A hip-dominant squat has less peak knee flexion (Chiu, vonGaza, & Jean, 2017, Fry, Smith, & Schilling, 2003, Kernozek et al., 2018, List et al., 2013, Lorenzetti et al., 2012) and less peak ankle dorsiflexion (Chiu, vonGaza, & Jean, 2017, Fry, Smith, & Schilling, 2003, Kernozek et al., 2018, List et al., 2013) compared to a knee-dominant squat. In addition, a hip-dominant squat has greater peak trunk flexion when squat depth is controlled for between squat types (Fry, Smith, & Schilling, 2003). These kinematic differences not only characterize each squat type, but they also contribute to differences in joint kinetics. A hip-dominant squat has a larger peak internal hip extension moment and a smaller peak internal knee extension moment compared to a knee-dominant squat (Chiu, vonGaza, & Jean, 2017, Fry, Smith, & Schilling, 2003, Kernozek et al., 2018, List et al., 2013, Lorenzetti et al., 2012). For these reasons, clinicians and coaches may select a hip-dominant squat to include more focused strengthening of the hip and trunk extensors. Hip-dominant movement patterns also reduce patellofemoral joint force and stress (Escamilla, Rafael et al., 2009, Escamilla, Rafael F. et al., 2008, Hofmann, Holyoak, & Juris, 2017, Kernozek et al., 2018), patellar tendon force (Scattone Silva et al., 2017), and anterior cruciate ligament force and strain (Escamilla, Rafael F. et al., 2012, Kulas, Hortobagyi, & DeVita, 2012). During the early stages of rehabilitation for a knee injury, it is common for a clinician to prescribe a hip-dominant squat to encourage knee mobility, knee joint loading, and functional movement while managing knee joint inflammation and pain.

Based on clinical experience and biomechanical concepts, it is generally accepted that clinicians may prescribe a knee-dominant squat for more focused knee extensor

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strengthening and for progressive knee joint loading. This type of squat may also be used to improve a patient's tolerance to loading the knee in a functional position where the knee passes in front of the individual's toes. This position is often uncomfortable for patients with knee pain, but is essential for daily activities (e.g., stair descent) and sport-specific activities.

For both squat types, instruction often involves specific cueing focused on anterior knee position to drive kinematic and kinetic changes. Literature has demonstrated a correlation between anterior knee position and trunk position during the bilateral squat (Fry, Smith, & Schilling, 2003) and other tasks (Farrokhi et al., 2008, Scattone Silva et al., 2017). However, it is less clear if trunk position during a bilateral squat has similar effects on joint kinematics and kinetics as knee position. A small, within-subject study (N = 16) by Straub et al. (2021) demonstrated that the knee extension moment increased with increased shank inclination and with decreased trunk angle in a weighted bilateral squat. They did not, however, include hip extension moment and only analyzed two squat depths (60° and 90° of knee flexion). The purpose of our study was to evaluate the effect of self-selected sagittal plane trunk position and knee position on the distribution of joint moments between the hip and knee extensors at multiple depths during the body-weight bilateral squat. We hypothesized that increased trunk flexion angle would correlate with increased hip extension moment and / or decreased knee extension moment as indicated by an increase in the ratio between the two joint moments (hip-to-knee moment ratio). We also hypothesized that sagittal plane trunk angle would have a greater impact on joint moments than shank angle, our proxy for anterior knee position.

2. METHODS:

2.1 Participants:

Data in this study were collected from participants recruited between November 2012 and November 2017 for a larger study on movement patterns in individuals with and without hip pain. While not the focus of this analysis, more information about the participants with and without hip pain can be found in Lewis et al. (2021). All participants had to be between the ages of 14 and 50 years. Exclusion criteria for both groups included history of neurological disorder, back or lower extremity surgery, or back or lower extremity (except for the hip) pain over the last two months. One hundred participants with kinematic and kinetic data for the bilateral squat were included in this study (Table 1). Gait data from an overlapping data set have been published elsewhere (Lewis et al., 2021).

This research was approved by the Institutional Review Boards of Boston University and Boston Children's Hospital. All participants provided written informed consent prior to participation in the study.

2.2 Instrumentation:

Kinematic data were collected using a 10-camera motion capture system (Vicon Motion Systems Ltd., Centennial, CO, USA). Each camera recorded the movement of 42 reflective markers that were positioned over bony landmarks on the participant's trunk, pelvis, and lower extremities. Specifically, markers were placed over the seventh cervical vertebra,

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acromion process, xiphoid process, iliac crest, anterior superior iliac spine (ASIS), sacrum, greater trochanter, lateral femoral epicondyle, medial femoral epicondyle, lateral malleolus, medial malleolus, first metatarsal head, fifth metatarsal head, and calcaneus bilaterally. An additional marker was placed over the right scapula to indicate the right side of the body. Cluster marker sets were used on the participant's thighs and shanks bilaterally (Cappozzo et al., 1997). Each set consisted of four reflective markers on a semi-rigid surface. Individual markers were secured to the participant with tape, and cluster marker sets were secured with felt wrap. Ground reaction force data were collected using force plates in a split-belt instrumented treadmill (Bertec Corp., Columbus, OH). Kinematic data and ground reaction force data were collected at a sampling rate of 100 Hz and 1000 Hz, respectively.

2.3 Procedures:

Participants were instructed to wear a form-fitting shirt, form-fitting shorts, and exercise shoes. Reflective markers were placed on the participant as described above. A static standing calibration trial was recorded to create a participant-specific model in Visual3D (C-Motion Inc., Germantown, MD, USA). For this trial, the participant was instructed to stand with feet shoulder-width apart and arms out to the side. Reflective markers on the medial epicondyles and the medial malleoli were removed following the calibration trial to allow for less restricted movement during the data collection.

Participants were instructed to begin the bilateral squat movement in a standing position with their feet shoulder-width apart and their arms out in front of them at a 90º angle from their trunk. Participants were then instructed to squat down in a controlled manner until their thighs were horizontal with the floor, or as close to that as could be achieved, and return to the standing position. Participants were not instructed on trunk position, knee position, or foot alignment. Participants performed practice trials if desired, then performed five repetitions of the bilateral squat, pausing in the start position before performing each squat. A trial was re-collected or excluded if performance was affected by loss of balance, discontinuous movement, or incomplete movement.

2.4 Data Processing:

Kinematic marker data were labelled in Vicon Nexus and processed in Visual3D. Marker trajectories were low-pass filtered using a fourth-order Butterworth filter with a 6 Hz cutoff frequency (Robertson and Dowling, 2003). Kinematic data were calculated using an eight-segment hybrid model with a CODA pelvis (Bell, Brand, & Pedersen, 1989) and a Cardan x-y-z (mediolateral, anteroposterior, vertical) rotation sequence (Cole et al., 1993). Hip, knee, and ankle joint angles were calculated as the angle between proximal and distal segments, and trunk and shank segment angles were calculated with respect to the global coordinate system. Ground reaction force data were low-pass filtered using a fourth-order Butterworth filter with a 10 Hz cutoff frequency (Robertson and Dowling, 2003). Kinematic marker data and ground reaction force data were used to calculate internal joint moments.

2.5 Data Analysis:

Variables of interest included internal hip extension and knee extension moments and the ratio between the two, defined as the hip-to-knee moment ratio. The hip-to-knee moment

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ratio was studied to provide a broader understanding of the relationship and distribution of moments between the hip and knee extensors during squatting activity. Trunk and shank segment angles were also evaluated. The hip-to-knee moment ratio was investigated in relation to the trunk and shank segment angles to determine the contribution of each segment angle to changes in the hip-to-knee moment ratio during the bilateral squat. Angle and moment data were extracted and the analysis performed at 6 squat depths, as determined by knee flexion angle, during the descent phase; the 6 analysis points were in 10° increments between 60° and 110° of knee flexion. We did not conduct analyses at points greater than 110° as less than 50% of our participants achieved 120° of knee flexion during the bilateral squat task. The average (standard deviation) peak knee flexion angle was 107° (16°).

2.6 Statistical Analysis:

To investigate the contribution of the sagittal plane trunk and shank angles to the hip-to-knee moment ratio, we used a linear regression model. The model included both angles and the ratio at the 6 knee flexion angles (KFAngle), representing squat depth, with data for the left and right sides for all participants (N = 100). Some participants did not achieve 110 degrees of knee flexion. These are included in the analysis as binary variables for each side (Achieves 110 left, right), denoted as 1 if they did not achieve 110 degrees and 0 if they did. Similarly, Side is a categorical variable with two levels representing each side. Controlling for all of these variables, we analyzed the effects of shank and trunk angle on Y, the hip-to-knee moment ratio, in the linear regression model:

$$\begin{aligned} Y = \beta_0 + \beta_1 TrunkAngle + \beta_2 ShankAngle + \beta_3 KFAngle + \beta_4 Side + \beta_5 Side \\ * TrunkAngle + \beta_6 Side * ShankAngle + \beta_7 Achieves110left \\ + \beta_8 Achieves110right \end{aligned}$$

This model included the interaction terms to study whether the relationship between shank and trunk angles varied with each side. To analyze the relative importance of the shank angle and trunk angle variables through their contributions to the proportion of variance explained by them, we (1) compared the above model with the models including only shank and trunk angles and without the trunk angle using Akaike Information Criteria (AIC) and the predictive power of the models, and (2) used the *Lindeman, Merenda, Gold* (lmg) relative importance technique (Lindeman, Merenda, & Gold, 1980). The lmg relative importance technique, implemented in the R package *relaimpo* developed by Groemping (2006), is used to estimate the proportion of variance explained by each variable; correlations among predictors does not affect the calculation of relative importance. All analyses were conducted in R.

3. RESULTS:

Average and standard deviation (SD) for hip-to-knee moment ratio, trunk angle, and shank angle data during the descent phase of the bilateral squat are presented in Table 2 for the 1021 observations (1 per side for each participant at each squat depth achieved).

We found the linear model to be a good fit explaining 62.21% of the variance, yielding small but highly significant effects for all three ratio predictors (trunk angle, shank angle,

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and squat depth, p <.001, Table 3). The model satisfies the conditions required for linear regression, and any corrections for heteroskedasticity do not alter the significance or the coefficients. The hip-to-knee moment ratio increased with increasing trunk angle and squat depth, and decreased with increasing shank angle. There was no significant effect or interaction of side (p≥.175), and no effect of achieving a greater peak squat depth (p≥.282).

The difference in the impacts of trunk and shank angles is apparent in not just the direction, but also the magnitude. Controlling for all other variables, both angles are significant (p<.001); a one degree increase in trunk angle increased the hip-to-knee moment ratio by 2% and a one degree increase in shank angle decreased the ratio by 2.9%.

Based on AIC, we found that the models containing all the variables had a reasonably good predictive power as compared to simpler models, particularly the ones without trunk angle. We also ranked the predictors in terms of their relative importance in the model by computing the proportion of the total variance that each variable was able to explain in the model. We found that trunk angle was the highest contributor of all, explaining 35% of the variance, and hence this establishes its significance as a predictor in the model (Figure 2)

4. DISCUSSION:

Our findings revealed that sagittal plane trunk angle and shank angle each contributed to the prediction of the hip-to-knee moment ratio during the bilateral squat. While trunk angle and shank angle each were significant, trunk angle demonstrated greater importance, explaining a larger proportion of the variance than shank angle. These findings support the idea that trunk position impacts hip and knee moments, while also acknowledging the influence of shank angle, our proxy for anterior knee position, on the moment ratio.

To our knowledge, this study is the first of its kind to investigate, in a large cohort, the influence of both trunk angle and shank angle on hip and knee moments during a self-selected body-weight bilateral squat. This study also compared the contribution that each of these segments has on the hip-to-knee moment ratio. Our finding that sagittal plane shank angle contributes to hip and knee moments during the bilateral squat, and decreases the hip-to-knee moment ratio when increased, is consistent with other studies (Chiu, vonGaza, & Jean, 2017, Fry, Smith, & Schilling, 2003, Kernozek et al., 2018, List et al., 2013, Lorenzetti et al., 2012, Straub et al., 2021).

Current literature lacks investigation into the effects of sagittal plane trunk position on lower extremity kinematics and kinetics during the bilateral squat as only one study has explored this relationship (Straub et al., 2021). Similar to our findings, this study demonstrates that trunk angle and shank angle each influences knee extension moments during the bilateral squat. While their study investigated this relationship in weighted squats, we found this relationship in squats with only body-weight. As with our study, they reveal that performing a bilateral squat with greater forward trunk flexion causes the knee extension moment to decrease. This study also reports the larger influence that sagittal plane trunk angle has on the knee extension moment compared to shank angle (Straub et al., 2021). No analysis was performed for hip extension moment. The influence of trunk mechanics has been explored

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in other tasks, including a forward stepping lunge (Farrokhi et al., 2008) and drop jump landings (Scattone Silva et al., 2017, Shimokochi et al., 2013). These studies reveal that movements performed with a forward flexed trunk compared to an extended trunk have a significantly lower knee extension moment (Scattone Silva et al., 2017, Shimokochi et al., 2013) or impulse (Farrokhi et al., 2008) and a significantly higher hip extension moment (Shimokochi et al., 2013) or impulse (Farrokhi et al., 2008), which is consistent with the findings of our current study.

Biomechanical principles can help explain our findings. When an individual performs a bilateral squat with a hip-dominant or knee-dominant movement pattern, their mechanics influence the magnitude of moment produced around each joint. If an individual performs a squat with less anterior knee movement and more forward trunk flexion (i.e. hip-dominant movement pattern), the mass of the trunk is farther forward from the hips and closer to the knees, creating a larger hip moment and smaller knee moment; therefore, the hip-to-knee moment ratio is larger with a larger trunk angle. If an individual performs a squat with greater anterior knee movement and a more upright trunk (i.e. knee-dominant movement pattern), the mass of the trunk has a smaller moment arm anterior to the hip joint and larger moment arm posterior to the knee joint, decreasing demand from the hip extensors and increasing demand from the knee extensors, respectively, reducing the hip-to-knee moment ratio.

Our findings suggest that trunk position may be used to instruct squat technique with the goal of modifying hip and knee extension moments, thereby providing clinicians and coaches with more options for cueing and instruction. For example, instructing an individual to squat with a greater forward trunk angle by hinging at the hips may encourage more of a hip-dominant movement pattern, while instructing an individual to squat with a more upright or vertical trunk may encourage more of a knee-dominant movement pattern. A variety of cueing options may provide patients with more opportunity for success in performing a particular movement pattern as individuals may find greater ease understanding and acting upon one cue versus another depending on their pain, motor control, and / or cognition. Providing a cue for trunk position may also be uniquely beneficial for groups with chronic knee pain and those undergoing post-operative care for the knee, both of which commonly demonstrate compensatory strategies to decrease the amount of loading at the knee (Orishimo et al., 2010, Roos, Button, & van Deursen, 2014, Scattone Silva et al., 2017, Sigward et al., 2018, Wren et al., 2018). These individuals may also better respond to a cue directed at trunk position due to attentional focus being placed on an area of the body other than their involved knee.

One limitation of our study was that we investigated the relationship between self-selected trunk position and lower extremity joint moments, limiting our ability to generalize our findings to explain how modifying trunk position may affect hip and knee moments. In addition, our study investigated body-weight squats performed on level ground; these findings may not transfer to other tasks, such as variations of the bilateral squat (e.g., weighted squats, heels elevated) or single leg squats. This study included individuals with hip pain. As our primary focus was on the biomechanical relationship between trunk and shank angles and the hip-to-knee moment ratio, we did not assess the effect of hip pain on

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angles or moments, but did confirm that our results were similar when including the effect of group (hip pain vs. no pain) in the model. This study included primarily young adults (average age of 25.5 years), limiting our ability to generalize our findings to older adults.

5. Conclusion

This study evaluated the distribution of joint moments between the hip and knee extensors in a large cohort of participants at multiple points during the bilateral squat. The bilateral squat was performed with a self-selected squat pattern. While clinicians typically focus on knee position to alter hip and knee moments during the squat, this study demonstrated that knee position and trunk position each affects hip and knee moments during the movement. Therefore, trunk position may be used to instruct squat technique with the goal of modifying hip and knee moments. Future research should investigate how modifying trunk position affects the hip-to-knee moment ratio.

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This research was approved by the Institutional Review Boards of Boston University and Boston Children's Hospital. All participants provided written informed consent prior to participation in the study.

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Highlights

- The bilateral squat is commonly used for rehabilitation and strengthening.
- Trunk and shank angle each contribute to the hip-to-knee moment ratio.
- Trunk position demonstrates a larger influence on the hip-to-knee moment ratio.
- Trunk position may be used to instruct squat technique to modify hip and knee moments.

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Figure 1:
Depiction of hip-dominant (light gray) and knee-dominant (dark gray) squat positions at the same degree of knee flexion. The hip-dominant squat, with the knees staying behind the toes, better incorporates the hip extensors. The knee-dominant squat, with the knees moving anterior to the toes, primarily targets the knee extensors. While the two squats differ in knee position, the squats also differ in trunk position.

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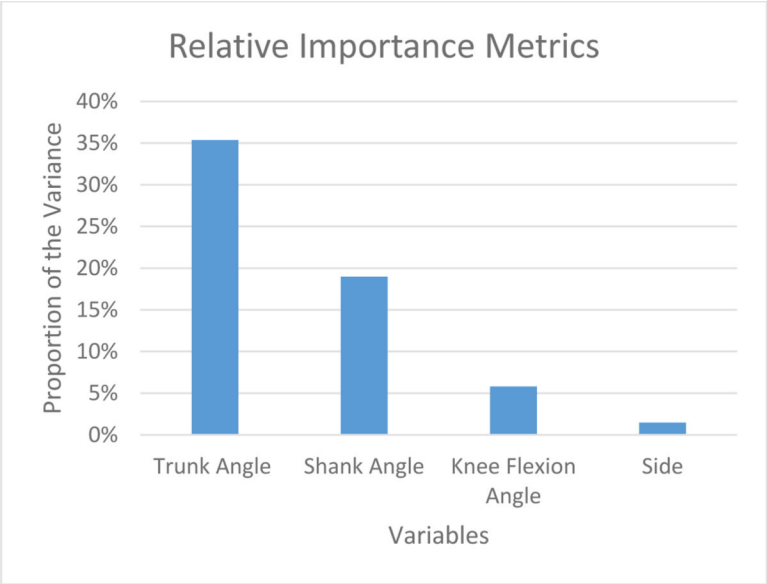

Figure 2:

Relative importance metrics for the variables explaining 1% or more of the variance in the hip-to-knee moment ratio.

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Table 1:

Participant characteristics.

Self-reported sex	66 females, 34 males
Age, years (mean ± SD)	25.5 ± 8.1
	(Range: 14 – 18 years)
Height, m (mean ± SD)	1.71 ± 0.10
Mass, kg (mean ± SD)	68.7 ± 13.0

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Table 2:

Average and standard deviation (SD) for hip-to-knee moment ratio, trunk angle, and shank angle data at each knee flexion angle analyzed during the descent phase of the bilateral squat.

Knee Flexion Angle	N	Hip-to-Knee Moment Ratio		Trunk Angle		Shank Angle	
		Average	SD	Average	SD	Average	SD
60°	200	0.69	0.44	23.4	8.9	23.5	4.6
70°	200	0.73	0.40	27.1	9.2	26.4	5.0
80°	195	0.78	0.38	30.3	9.4	29.1	5.5
90°	181	0.82	0.37	33.0	9.7	31.8	5.9
100°	145	0.83	0.34	35.4	9.6	34.0	6.1
110°	100	0.88	0.32	37.8	9.7	35.6	6.7

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Table 3:

Linear regression model summary with the hip-to-knee moment ratio as the dependent variable.

	Estimate	Std Error	t value	P-value
(Intercept)	0.813	0.142	5.722	<.001
Trunk Angle	0.020	0.002	8.042	<.001
Shank Angle	−0.030	0.004	−7.698	<.001
Knee Flexion Angle	0.063	0.009	7.047	<.001
Side	−0.048	0.089	−0.536	.592
Achieves 110 left	−0.040	0.037	−1.077	.282
Achieves 110 right	0.013	0.037	0.352	.725
Trunk Angle*Side	−0.002	0.001	−1.359	.175
Shank Angle*Side	0.000	0.002	0.192	.848